

CSE 400: Fundamentals of Probability in Computing

School of Engineering and Applied Sciences, Ahmedabad University

CSE 400 – Tutorial 2

Exam Preparation Lecture Scribe

Q1. Random Point on a Line Segment

Problem

A point is chosen at random on a line segment of length L . Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than $\frac{1}{4}$.

Concept Used

- Uniform distribution on an interval

Step-by-Step Reasoning

Let the random point be at distance X from one endpoint.

Since the point is chosen at random on a segment of length L :

$$X \sim \text{Uniform}(0, L)$$

The two segments formed are:

$$X \quad \text{and} \quad L - X$$

The shorter segment is:

$$\min(X, L - X)$$

The longer segment is:

$$\max(X, L - X)$$

We require:

$$\frac{\min(X, L - X)}{\max(X, L - X)} < \frac{1}{4}$$

Case 1: $0 < X \leq \frac{L}{2}$

Then:

$$\min(X, L - X) = X$$

$$\max(X, L - X) = L - X$$

Condition becomes:

$$\frac{X}{L - X} < \frac{1}{4}$$

Multiply both sides by $(L - X)$:

$$X < \frac{1}{4}(L - X)$$

$$X < \frac{L}{4} - \frac{X}{4}$$

$$X + \frac{X}{4} < \frac{L}{4}$$

$$\frac{5X}{4} < \frac{L}{4}$$

$$5X < L$$

$$X < \frac{L}{5}$$

Case 2: $\frac{L}{2} < X < L$

Then:

$$\min(X, L - X) = L - X$$

$$\max(X, L - X) = X$$

Condition becomes:

$$\frac{L - X}{X} < \frac{1}{4}$$

$$L - X < \frac{X}{4}$$

$$L < \frac{5X}{4}$$

$$4L < 5X$$

$$X > \frac{4L}{5}$$

Favourable Region

$$X < \frac{L}{5}$$

or

$$X > \frac{4L}{5}$$

Total favourable length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Total possible length = L

Probability

$$P = \frac{\frac{2L}{5}}{L}$$

$$P = \frac{2}{5}$$

Q2. Normal Distributions and Error Probability

Concepts Used

- Normal distribution
- Bayes' classification principle
- Conditional probability

White region:

$$X \sim N(4, 4)$$

Black region:

$$X \sim N(6, 9)$$

Fraction black = α

Given reading = 5

Error probability equal means:

$$P(\text{white}|5) = P(\text{black}|5)$$

Using Bayes' rule:

$$\alpha f_B(5) = (1 - \alpha) f_W(5)$$

Compute densities:

$$\begin{aligned} f_W(5) &= \frac{1}{\sqrt{2\pi(4)}} e^{-\frac{(5-4)^2}{2(4)}} \\ &= \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{8}} \\ f_B(5) &= \frac{1}{\sqrt{2\pi(9)}} e^{-\frac{(5-6)^2}{18}} \\ &= \frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{18}} \end{aligned}$$

Thus:

$$\alpha \frac{1}{3\sqrt{2\pi}} e^{-1/18} = (1 - \alpha) \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Solve for α .

Q3. Minimizing $E|X - a|$

(a) Uniform on $(0, A)$

Concept: Uniform distribution

$$X \sim U(0, A)$$

$$E|X - a| = \int_0^A |x - a| \frac{1}{A} dx$$

Split at a :

$$= \frac{1}{A} \left[\int_0^a (a-x)dx + \int_a^A (x-a)dx \right]$$

Compute each integral step by step.

Minimization yields:

$$a = \frac{A}{2}$$

(b) Exponential with rate λ

Concept: Exponential distribution

$$f(x) = \lambda e^{-\lambda x}$$

$$E|X - a| = \int_0^\infty |x - a| \lambda e^{-\lambda x} dx$$

Split at a .

Minimization yields:

$$a = \frac{\ln 2}{\lambda}$$

Q4. Exponential Mixture

Concepts:

- Exponential distribution
- Conditional probability

$$P(T > t + s | T > t) = \frac{P(T > t + s)}{P(T > t)}$$

Mixture survival function:

$$P(T > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

Thus:

$$\frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Independent Poisson Variables

Concept:

- Poisson distribution
- Independence

$$P(X = k | X + Y = n) = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Using Poisson formula:

$$\frac{e^{-\lambda_1} \frac{\lambda_1^k}{k!} \cdot e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!}}{e^{-(\lambda_1+\lambda_2)} \frac{(\lambda_1+\lambda_2)^n}{n!}}$$

Simplify:

$$\binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Q6. Transformation $Y = g(X)$

Concept:

- Uniform distribution
- CDF definition
- PDF as derivative of CDF

$$F_Y(y) = P(Y \leq y)$$

Compute using graphical mapping of intervals.

Then:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Q7. Stock Movement Model

Concept:

- Independent Bernoulli trials
- Binomial approximation

After 1000 periods:

$$\text{Price} = su^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \geq 1.3$$

Take log:

$$k \ln u + (1000 - k) \ln d \geq \ln 1.3$$

Solve for k .

Probability:

$$P(K \geq k_0) = \sum_{k=k_0}^{1000} \binom{1000}{k} p^k (1-p)^{1000-k}$$

Q8. Exponential Distribution

$$\lambda = \frac{1}{2}$$

(a)

$$P(T > 2) = e^{-\lambda 2} = e^{-1}$$

(b)

$$P(T > 10 | T > 9) = \frac{e^{-\lambda 10}}{e^{-\lambda 9}} = e^{-\lambda} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

Concept:

- Joint pdf of independent variables
- Transformation

Joint pdf:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

$$\int \int_{x^2+y^2 \leq z^2} f_X(x)f_Y(y) dx dy$$

Differentiate:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Q10. Joint Distribution of M_n and M_{n+1}

Concept:

- Maximum of i.i.d. random variables

$$M_n = \max(X_1, \dots, X_n)$$

$$P(M_n \leq x) = F(x)^n$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

Joint distribution:

$$P(M_n \leq x, M_{n+1} \leq y)$$

Case analysis:

If $y < x$: probability 0.

If $y \geq x$:

$$P(M_n \leq x, X_{n+1} \leq y) = F(x)^n F(y)$$

End of Tutorial-2 Exam Scribe